N95 Day 2016: Proper Use, Filtration, and Fit – The Three-Legged Stool of Respiratory Protection

September 6, 2016 by Jaclyn Krah and Ronald Shaffer



Dear N95 Day enthusiast – We want to thank you for joining us for another year of N95 respiratory protection information celebration. We know – today is 9/6, not 9/5. But we hope that many of you enjoyed the Labor Day holiday yesterday and are rested and ready for our annual N95 Day holiday today! NIOSH established this observance because N95 respirators are so important to the health and safety of workers in so many different industries. Much of the information this year focuses on the big three factors that contribute to effective respiratory protection: proper respirator use, filtration, and fit.

These three factors are the three structural legs of respiratory protection. The analogy of the three-legged stool has been applied in popular literature for decades to describe retirement benefits, healthcare, and business models, among other topics. We're flexing our metaphorical muscles and adding respiratory protection to that list because the analogy just works so darn well. You don't have to be an engineer to know that a three-legged stool requires legs of equal length – otherwise you get that annoying, unstable wobble.

To a respirator user whose workplace requires the N95 level of protection, the three components of proper respiratory protection are even more important. The protection a user receives from a properly worn N95 filtering facepiece respirator or elastomeric respirator with an N95 cartridge (for the purpose of this blog, we will combine these two into the general term "N95 respirator") can have a direct effect on that user's health and safety. To be effective, respiratory protection requires three main things: (1) the respirator has to be put on correctly and worn during the exposure; (2) the respirator needs to capture 95% or more of the particles that pass through the filter (where the 95 in N95 comes from); and (3) the respirator must fit the user's face to minimize the risk that particles will simply bypass the filter and get into the breathing zone through gaps between the user's skin and the respirator seal. If any one of these falls short, the wearer could be in store for bigger problems than an annoying wobble.

One fancy way of estimating exposure reduction considers two factors: wear-time and protection. (See references below.) Wear-time is the amount of time the respirator is worn properly when the hazard is present, while protection is a function of respirator fit and particle filtration.

Proper Use

Although it may seem obvious, it is worth repeating that if the respirator is not worn or used properly during exposure, it will not keep the wearer from coming into contact with hazardous particles. Proper respirator use has been an on-going theme of N95 Day and the NIOSH Science Blogs. For those in the healthcare industry, we are introducing our new Hospital Respiratory Protection Resources webpage to help you quickly find needed resources. Studies have shown that workers are more likely to comply with respirator requirements and best practices when they understand how and why the devices work. NIOSH has been developing tools to assist hospital respiratory protection program managers perform this important function. The new website will serve as a hub for these types of NIOSH resources and make it easier for hospital respiratory protection program managers and interested users to find both NIOSH-generated information that supports OSHA's 1910.134 respiratory protection standard, as well as more approachable education tools, including infographics, fact sheets, and training videos.

We've now established the first leg of our stool, but you don't get to sit on it quite yet…

Filtration

As outlined in 42 Code of Federal Regulations Part 84, NIOSH applies test conditions that represent near "worst case" scenarios when testing N95 respirators submitted for approval. **This ensures that filters in NIOSH-approved respirators will collect different workplace aerosols (including emerging hazards such as airborne infectious organisms and engineered nanoparticles) with high efficiency.** While the science of particle filtration has been known for decades, we still get questions regularly through our PPE hotline and CDC-info from users, policy-makers, and respiratory protection program managers about whether NIOSH-approved N95 respirators will filter out specific hazards. To help explain the concepts of particle filtration and address these questions, NIOSH developed an educational video titled "A Particle is a Particle."

Filtration makes up the second leg of the stool, but it's still not quite safe (balanced?)…

Fit

Respirator fit can be a complex subject. OSHA's 1910.134 respiratory protection standard requires respirator fit testing initially and annually thereafter. Within the last two decades, NIOSH has published influential research demonstrating why initial respirator fit testing is so important and providing additional scientific justification in support of annual fit testing. Our webinar this year is titled *"The Science behind Respirator Fit Testing in the Workplace: Past, Present, and Future."* **This webinar will help you answer two questions… Why do we have to fit test? And why every year?!** For more information about the webinar as well as other N95 Day resources, see the N95 Day webpage. If you missed the webinar, don't worry. It will be posted online after the close of the event. (This article will be updated with the new link.)

With these three components in place, we now have a fully-functional, safe and stable, NIOSH-approved three-legged stool. (Metaphorically speaking, of course. We don't "approve" furniture.)

We appreciate your participation in N95 Day, showing your dedication to occupational safety and health through following and sharing our social media on Twitter (@NIOSH, @NPPTL), Facebook, and Instagram (nioshusa). Please use today as an excuse to put aside the other distractions of your busy schedule and delve into our resources. Have a happy and safe N95 Day!

Additional Links to Explore

N95 2016 Infographics





Previous N95 Day Science Blogs

- 2012: Happy N95 Day! (aka, What is N95 Day?)
- 2013: A Guide to N95 Resources
- 2014: Respirator Preparedness – Where Technology Meets Good Practices
- 2015: The tools to build a culture of proper respiratory protection practices

Previous Science Blogs on N95 Respirator Use, Fit, and Filtration

- Reaching Towards a Healthier, Safer Workplace: NIOSH looks at healthcare worker familiarity with recommended respiratory protection practices
- N95 Respirators and Surgical Masks

- Respiratory Protection for Workers Handling Engineered Nanoparticles
- New NIOSH Study Supports the OSHA Annual Fit Testing Requirements for Filtering Facepiece Respirators

References

- Gosch ME, Shaffer RE, Eagan AE, Roberge RJ, Davey VJ, Radonovich LJ. B95: a new respirator for health care personnel. American journal of infection control. 2013 Dec 31;41(12):1224-30.
- Janssen L, Ettinger H, Graham S, Shaffer R, Zhuang Z. Commentary: the use of respirators to reduce inhalation of airborne biological agents. Journal of occupational and environmental hygiene. 2013 Aug 1;10(8):D97-103.

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Personal Protective Equipment, Respirators

7 comments on "N95 Day 2016: Proper Use, Filtration, and Fit – The Three-Legged Stool of Respiratory Protection"

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For further clarification – the new NIOSH ruling facial hair requirements about no hair will be between the sealing surface and face is applicable to the N95 series respirators, correct?
Reply

This is great information!
Reply

We use PPE everyday ,we are death and trauma cleaners. So we appreciative good personal protective gear. The N95 respirator mask is now even more important to us everyday even in between our crime scene cleanup jobs. We are [name removed] This new corona virus outbreak is making them more important to everyone. Everyone is learning about them, PPE and disinfection services. Great article.
Reply

I think all N95 and surgical mask should be laboratory tested after used by users in hospital s department such as icu/hdu (just after used in AGP environment}
To find out if the virus or bacteria really get to the masks.
Reply

Requesting protection clarification on the relationship between N95 &
Covid 19
————————————————
N95 respirator:

(1)The N95 respirator pore size = 0.3 micron (by 3M)

(2) N95s by default are protective of tuberculosis (bacteria size 3-5 micron).

(3) The Exhaled air from the "provider to the patient" has NO protection of protecting the patient(assuming the provider is infected by the COVID virus).
————————————————
Question:
(1) How would N95 respirator would be protective of COVID 19 ?
(Analyzing from the stand point of the pore size of the respirator and the size of COVID 19)

(2) The exhaled air from the (assumed infected) provider to patient has no protection: who is infecting who?

(3) Can N95 respirator technically offer any protection against
contracting COVID 19 at all, (from the stand point of the mask pore size and the virus size). Thanks
Warm Regards
Dr Karim
Cell: 617-240-0272
Reply

> Thank you for your comment. The answer to your question can be understood first by noting that main premise of your comment is not correct. You state, “The N95 respirator pore size = 0.3 micron (by 3M).” The reason this is incorrect is that N95 filters do not have a “pore” or a “pore size.” The filters are a weave of electrostatically charged fibers. They collect particles by: inertial impaction and interception mechanisms of larger particles, and diffusion of smaller particles. Once the particles contact the filter, they are captured. The N95 filters do not work like a strainer or sieve with pores or mesh holes of a particular diameter. This was described in detail in the 2009 NIOSH Science Blog: https://blogs.cdc.gov/niosh-science-blog/2009/10/14/n95/. The N95 filters are highly efficient at collecting small aerosol particles smaller and larger than 0.3 micrometers (microns, aerodynamic diameter). NOTE: the efficiency of collection by inertial impaction and interception drops as particles become smaller. The efficiency of collection by diffusion mechanism drops as particles become larger. The minimum for the combined collection efficiencies for inertial impaction and diffusion is somewhere between 0.1 – 0.3 micrometers, which is why NIOSH tests particulate respirators in its certification tests with particles in that most challenging size range, around 0.3 micrometers.
>
> The second assumption you make is that airborne viruses are floating in the air at the size observed for individual virions under an electron microscope. In fact, the individual virions are carried along inside larger droplets that include saliva, mucous, proteins, and other bodily fluids which makes the virus-containing aerosol larger than individual virions.
>
> You also state, “The exhaled air from the (assumed infected) provider to patient has no protection: who is infecting who?” Recent NIOSH test data show that N95 masks do provide source control in the event that the mask wearer is also infected.
>
> The bottom line is that an N95 respirator offers substantial protection against aerosols that may contain the SARS-CoV-2 virus which is the cause of COVID-19 disease, and they are effective as source control protecting others from the mask-wearer if that person is infected.

Thank you for this write up. N95 Day is an elegant way to promote occupational safety. Having trained/worked in mining, construction, mountain rescue and medical facilities, there are instances when the P100 type of resourator is more useful – not only because it is more effective at filtrarion, but also because it is washable, more durable and more resistant to wet hazards. Otherwise, keep up the great work..
Reply

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